

– TECHNICAL DATA SHEET

NN-395 Superplasticizer

High-Range Water-Reducing Concrete Admixture — SNF Based

Sulphonated naphthalene formaldehyde condensate · liquid superplasticizer to IS 9103



NN-395 — dark brown liquid SNF superplasticizer, supplied in 220 kg drums / bulk

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-06	1.0	1 July 2026	NN-395

01 PRODUCT DESCRIPTION

NN-395 is a liquid high-range water-reducing concrete admixture (superplasticizer) based on sulphonated naphthalene formaldehyde (SNF) condensate. It imparts a powerful plasticizing effect, allowing substantial water reduction to promote high ultimate strengths, or greatly increased workability at constant water content. NN-395 is chloride-free and does not attack reinforcement.

The product is intended for high-quality concrete under demanding conditions — high ambient temperature, long-haul ready-mix supply and wherever dense, durable, high-strength concrete is required. It conforms to the requirements of IS 9103 for superplasticizing admixtures.

02 USES

- **RMC** — Ready-mix concrete plants, especially where long haulage and workability retention are required.
- **High strength** — High-strength and high-performance concrete at low water/cement ratio.
- **Flowing concrete** — Pumped, flowing and difficult-to-place concrete in congested reinforcement.
- **Hot weather** — Hot-weather and long-transport concreting where slump retention matters.

03 CHARACTERISTICS & ADVANTAGES

- **Slump control** — Long-lasting control of slump loss for extended workability.
- **Water reduction** — Improved workability without increasing water; reduced water without loss of workability.
- **Higher strength** — Increased ultimate compressive and flexural strength.
- **Durability** — Denser concrete with improved water-tightness and surface finish.
- **Chloride-free** — Chloride-free — does not attack or corrode steel reinforcement.

04 PHYSICAL PROPERTIES

Property	Value
Chemical base	Sulphonated naphthalene formaldehyde (SNF) condensate
Appearance / colour	Dark brown liquid
Relative density	≈ 1.12 kg/l at 25 °C
Chloride content	Nil (chloride-free)
Physical state	Free-flowing liquid

05 PERFORMANCE REQUIREMENTS TO IS 9103

Superplasticizing (normal) type. Values relative to a control mix without admixture, per IS 9103 Table 1A / 1B.

Parameter	Standard
Water content (for equal workability)	≤ 80 % of control (≥ 12 % water reduction, min per IS 9103)
Compressive strength — 1 day	≥ 90 % of control
Compressive strength — 3 / 7 / 28 days	≥ 90 % of control
Compressive strength — 6 months / 1 year	≥ 90 % of control
Flexural strength — 3 / 7 / 28 days	≥ 90 % of control
Loss of workability on standing	Slump at 45 min ≥ control mix at 15 min
High-workability flow (test mix)	510 mm – 620 mm
Length change over control (28 d / 6 mo / 1 yr)	≤ 0.010 % (increase), max
Bleeding increase over control	≤ 5 %, max

06 DOSAGE

Recommended dosage is 0.2 % to 2.0 % by weight of cement. Higher dosages may be used where agreed by the engineer and consultant. The optimum dosage must be finalised by site trial mixes (per IS 10262), as it depends on the cement, aggregates and mix design. Shake well before use.

Parameter	Guidance
Typical dosage range	0.2 – 2.0 % by weight of cement
Optimum dosage	To be established by site trial
Overdosing	May cause bleeding, segregation or set retardation

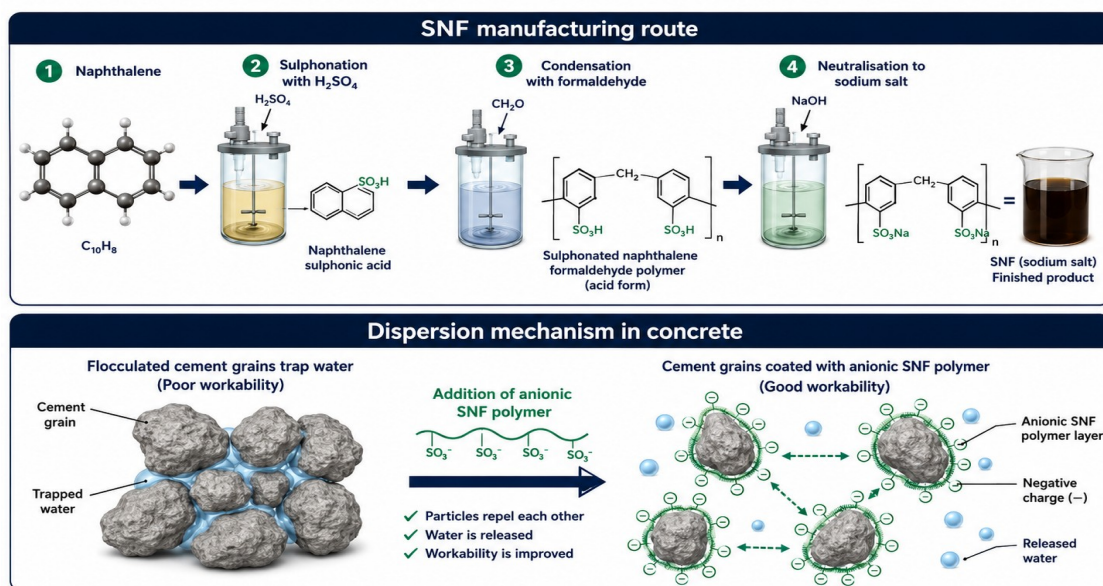
07 APPLICATION & DISPENSING



Liquid superplasticizer dosed into the mixing water at a ready-mix batching plant

- **Addition** — Add to the mixing water, or directly into the mixer after 50–70 % of the mixing water has been added. Do not add to dry aggregate or cement.
- **Mixing** — Ensure a minimum wet mixing time of about 60 seconds (depending on mixer and conditions) for full dispersion.
- **Practice** — Follow standard good-concreting practice for production, placing and testing; cure fresh concrete per the relevant standard.
- **Cleaning** — Clean tools and equipment with water immediately after use; hardened material must be removed mechanically.

08 MECHANISM & MANUFACTURING PROCESS



SNF manufacturing route and cement-grain dispersion mechanism

SNF is produced by sulphonating naphthalene, condensing it with formaldehyde and neutralising to the sodium salt. In concrete, the anionic SNF polymer adsorbs onto cement grains and imparts a strong negative charge; electrostatic repulsion disperses the flocculated particles, releasing trapped water and greatly increasing workability at a given water content – or allowing water to be reduced at constant workability.

09 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Packaging	220 kg drums / bulk supply
Storage	Store in dry conditions between +10 °C and +40 °C; protect from direct sunlight and frost
Shelf life	12 months from date of production in unopened, correctly stored packaging
If frozen / precipitated	May be used after slow thawing to room temperature and intensive re-mixing

10 COMPATIBILITY & LIMITATIONS

- **Compatibility** – May be used with other admixtures; always conduct laboratory and site trials before combined use.
- **Caution** – Do not mix different superplasticizer chemistries (e.g. SNF with PCE) on site – cross-reaction can cause slump loss.
- **Overdosing** – Excessive water addition or overdosing may cause bleeding or segregation.

All technical data are based on laboratory tests; actual measured values may vary due to circumstances beyond our control.

Disclaimer: *The information and recommendations relating to the application and end-use of this product are given in good faith based on current knowledge and experience when properly stored, handled and applied under normal conditions. As differences in materials, substrates and site conditions are beyond our control, no warranty of merchantability or fitness for a particular purpose is implied. Users must test the product's suitability for the intended application and comply with all applicable standards, including IS 9103 and IS 456.*